# Class #11 - Matrix Operations

1. Given $A = [a_{ij}] \in \mathbf{R^{m\times n}}$, $A$ *transpose* or *transpose of* $A$ is the matrix whose $i$th row is the $i$th column of the matrix $A$ for $i = 1, \cdots, m$.
   Notation: $A^T$

2. Example:$A = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$. $A^T =$ __________.

   Caution: $(0,1,0,0) \neq \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}$.

   Example: $A = \begin{bmatrix} 0 & 3 \\ 1 & -1 \\ 0 & 2 \end{bmatrix}$. $A^T =$ __________.

   Note: $(A^T)^T =$ __________.

3. Let $A, B \in \mathbf{R^{m\times n}}$. Then

   (i) $A + B = [a_{ij}] + [b_{ij}] = [c_{ij}]$ where $c_{ij} =$ __________. (add each individual entries)

   Note: To add two matrices, their sizes should be the same!

   (ii) $cA = [ca_{ij}]$. (scale each individual entries)

4. Note: $(A + B)^T =$ __________
   $(cA)^T =$ __________.

5. Multiplication of two matrices:
   Let $T(x) = Ax$ where $A \in \mathbf{R^{m\times n}}$, and $S(y) = By$ where $B \in \mathbf{R^{p\times m}}$.
   Note:

   (a) $T$ and $S$ are __________transformations.

(b) $S(T(x))$ is a *transformation* from __________ to __________.
Then, $S(T(x)) = B(Ax)$. Moreover,

$$B(Ax) = B(x_1a_1 + x_2a_2 + \cdots + x_na_n)$$
$$= B(x_1a_1) + B(x_2a_2) + \cdots + B(x_na_n) \quad \text{(why?)}$$
$$= x_1Ba_1 + x_2Ba_2 + \cdots + x_nBa_n \quad \text{(why?)}$$
$$= (BA)x$$

if we define the matrix multiplication $BA$ as

$$BA = \begin{bmatrix} Ba_1 & Ba_2 & \cdots & Ba_n \end{bmatrix}.$$

Note: $BA \neq AB$.

6. Example: $A = \begin{bmatrix} 1 & 3 & 6 \\ -1 & 5 & 0 \end{bmatrix}$ $\quad B = \begin{bmatrix} 2 & 3 \\ 1 & 1 \\ 0 & 3 \end{bmatrix}$ $\quad C = \begin{bmatrix} 0 & 5 & 1 \\ 3 & 4 & 1 \end{bmatrix}$

(a) $AB =$

Note: $(i, j)$-element of $AB$ = sum of componentwise product of __________ th __________ of $A$ and __________ th __________ of $B$.

(b) $BC =$

(c) $AC =$

7. Example: Let $I_n$ denote the $n \times n$ identity matrix, and let $A = \begin{bmatrix} 1 & 3 & 6 \\ -1 & 5 & 0 \end{bmatrix}$.

Fill in the blank: For $A \in \mathbf{R^{m \times n}}$, __________ $= A =$ __________.

8. True or False: Let $A, B$ be two nonzero matrices. Then $AB \neq 0$.

9. Theorem: Let $A, B$, and $C$ are matrices of appropriate sizes so that matrix multiplications are defined, and let $c$ be a scalar. Then

   (a) $A(BC) =$ __________

   (b) $A(B + C) =$ __________

   (c) $(B + C)A =$ __________

   (d) $c(AB) =$ __________ $=$ __________

10. Theorem: $(AB)^T =$ __________
    Proof: $(AB)_{ij} =$

    $((AB)^T)_{ji} =$ __________
    $(B^T A^T)_{ji} =$ sum of componentwise product of __________th row of __________and __________th column of __________
    $=$ sum of componentwise product of __________th __________of $B$ and __________th __________of $A$
    $= (AB)_{ij}$